Lecture Notes 5

Econometrics

Difference Equations

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From Cross-Sections to Time Series

Our earlier lectures treated observations as a random sample from a population. With time-series data, order matters: today depends on yesterday.

- Time-series econometrics is, at its core, the estimation of difference equations with stochastic components.
- A difference equation expresses a variable as a function of its own lagged values, time, and other variables.
- Uncovering this equation of motion lets us forecast: the predictable part of the series can be extrapolated forward.

This lecture is about how to solve such equations — the foundation for everything in time-series analysis.

Why Study Difference Equations? Forecasting

The original purpose of time-series analysis was forecasting – and forecasting works only because series are *persistent*: the past carries information about the future.

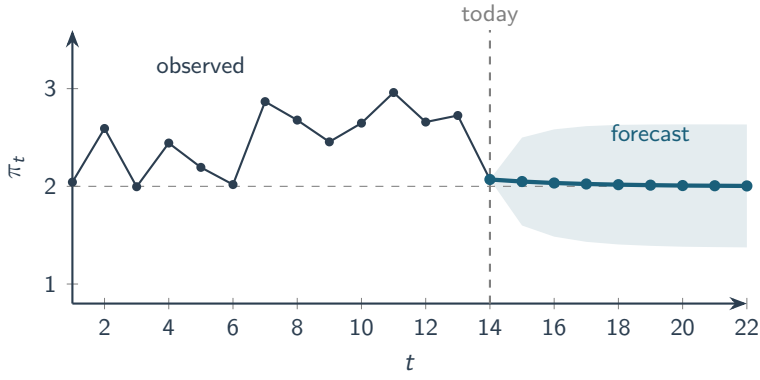
- **Example: inflation.** A high-inflation quarter tends to be followed by another. We can capture this with

$$\pi_t = a_0 + a_1 \pi_{t-1} + \varepsilon_t, \qquad 0 < a_1 < 1.$$

- This is a difference equation: today's value depends on its own past plus a shock.
- Knowing it, the best guess for next quarter is $\mathbb{E}(\pi_{t+1} \mid \pi_t) = a_0 + a_1 \pi_t$ — we *extrapolate* the predictable part forward.
- Looking further ahead, the forecast gradually reverts to a long-run average at a rate governed by a_1.

To forecast a series, we uncover its equation of motion and solve it.
That is exactly what this lecture teaches.

Why Study Difference Equations? Forecasting



A Motivating Example: Decomposing a Series

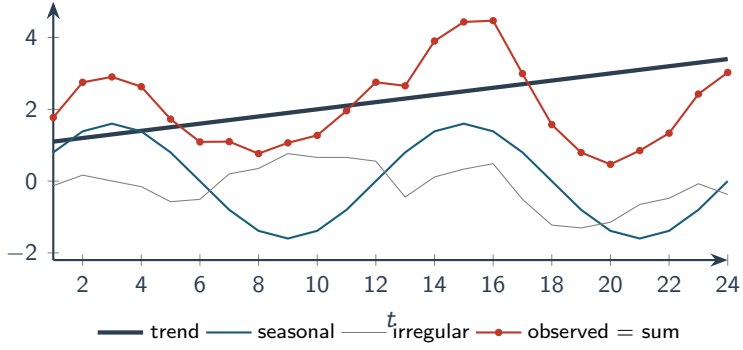
A classic view decomposes a series into **trend**, **seasonal**, and **irregular** components. For example:

$$\begin{aligned} \text{Trend:} \quad & T_t = 1 + 0.1\, t \\ \text{Seasonal:} \quad & S_t = 1.6 \sin(t\pi/6) \\ \text{Irregular:} \quad & I_t = 0.7\, I_{t-1} + \varepsilon_t \end{aligned}$$

- Each is a difference equation: the trend and seasonal depend on time; the irregular depends on its own lag plus a shock ε_t.
- The irregular part is partly predictable – large values cluster – but reverts to zero, since past shocks decay at rate 0.7^i.
- Forecasting the series means forecasting each component and summing.

A Motivating Example: Decomposing a Series

$$T_t = 1 + 0.1\,t, \qquad S_t = 1.6\sin(t\pi/6), \qquad I_t = 0.7\,I_{t-1} + \varepsilon_t$$



Economic Models as Difference Equations

Many theories have a natural representation as a stochastic difference equation — often with *testable* restrictions.

- **Random walk hypothesis.** Stock (log) prices should satisfy $y_{t+1} = y_t + \varepsilon_{t+1}$. In the more general form
$$\Delta y_{t+1} = \alpha_0 + \alpha_1 y_t + \varepsilon_{t+1},$$
the hypothesis is the restriction $\alpha_0 = \alpha_1 = 0$: rejecting it rejects the theory.
- **Multiplier–accelerator (Samuelson).** A Keynesian system in y_t, c_t, i_t collapses to a single second-order equation for GDP:
$$y_t = a\, y_{t-1} + b\, y_{t-2} + x_t.$$

The second-order case is exactly why we need solution methods beyond simple iteration.

The nth-Order Linear Difference Equation

The workhorse form throughout time-series analysis is

$$y_t = a_0 + \sum_{i=1}^{n} a_i \, y_{t-i} + x_t$$

- The order n is the longest lag of y that appears.
- Linear because every y_{t-i} enters to the first power; the a_i are treated as parameters.
- x_t is the forcing process – it can be a function of time, other variables, and/or stochastic shocks.
- An important special case for the $\{\varepsilon_t\}$ sequence is $x_t = \sum_{i=0}^{\infty} \beta_i \varepsilon_{t-i}$.

A key special case: set $x_t = \varepsilon_t$ (a random shock), giving the autoregression

$$y_t = a_0 + a_1 y_{t-1} + \ldots + a_n y_{t-n} + \varepsilon_t.$$

With $n = 1$, $a_0 = 0$, $a_1 = 1$, this is the random walk $y_t = y_{t-1} + \varepsilon_t$.

What Does It Mean to "Solve" a Difference Equation?

- A solution expresses y_t as a function of time t, the forcing process $\{x_t\}$, and any initial conditions – with *no* lagged y's on the right.
- A solution is a function, not a number: substituting it into the equation must yield an identity for all admissible t.
- Solutions need not be unique – they typically involve an arbitrary constant, pinned down later by an initial condition.

Solution vs. Reduced Form

Writing $I_t = 0.7I_{t-1} + \varepsilon_t$ as $I_t = 0.49I_{t-2} + \varepsilon_t + 0.7\varepsilon_{t-1}$ is still just a reduced form – it contains an unknown lagged y. A solution eliminates all lagged y's:

$$I_t = \sum_{i=0}^{\infty} (0.7)^i \varepsilon_{t-i}.$$

Solution by Iteration

If the value of y in some specific period is known, a direct method of solution is to iterate forward from that period to obtain the subsequent time path of the entire y sequence.

Take the first-order equation $y_t = a_0 + a_1 y_{t-1} + \varepsilon_t$ with a known initial value y_0. Iterating forward:

$$\begin{aligned} y_1 &= a_0 + a_1 y_0 + \varepsilon_1 \\ y_2 &= a_0 + a_0 a_1 + a_1^2 y_0 + a_1 \varepsilon_1 + \varepsilon_2 \\ &\vdots \end{aligned}$$

Repeating t times gives the general pattern

$$y_t = a_0 \sum_{i=0}^{t-1} a_1^i \; + \; a_1^t \, y_0 \; + \; \sum_{i=0}^{t-1} a_1^i \, \varepsilon_{t-i}.$$

- This is a genuine solution: only t, y_0, and the shocks appear.
- The shock ε_{t-i} enters with weight a_1^i — so $|a_1|$ governs how fast the past fades.

Iteration Without an Initial Condition

Without a known y_0, iterate *backward* instead. Repeatedly substitute $y_{t-1} = a_0 + a_1 y_{t-2} + \varepsilon_{t-1}$, and so on:

$$\begin{aligned} y_t &= a_0 + a_1 y_{t-1} + \varepsilon_t \\ &= a_0(1 + a_1) + a_1^2 y_{t-2} + \varepsilon_t + a_1 \varepsilon_{t-1} \\ &= a_0(1 + a_1 + a_1^2) + a_1^3 y_{t-3} + \varepsilon_t + a_1 \varepsilon_{t-1} + a_1^2 \varepsilon_{t-2} \\ &\vdots \\ &= a_0 \sum_{i=0}^{m} a_1^i + a_1^{m+1} y_{t-m-1} + \sum_{i=0}^{m} a_1^i \, \varepsilon_{t-i}. \end{aligned}$$

Now let $m \to \infty$. If $|a_1| < 1$:

- the lagged term $a_1^{m+1} y_{t-m-1} \to 0$ — the distant past is forgotten;
- the geometric sum $\sum_{i=0}^{m} a_1^i \to 1/(1 - a_1)$.

$$y_t = \frac{a_0}{1 - a_1} + \sum_{i=0}^{\infty} a_1^i \, \varepsilon_{t-i}$$

Iteration and Stability

The solution just derived shows the behavior hinges entirely on a_1:

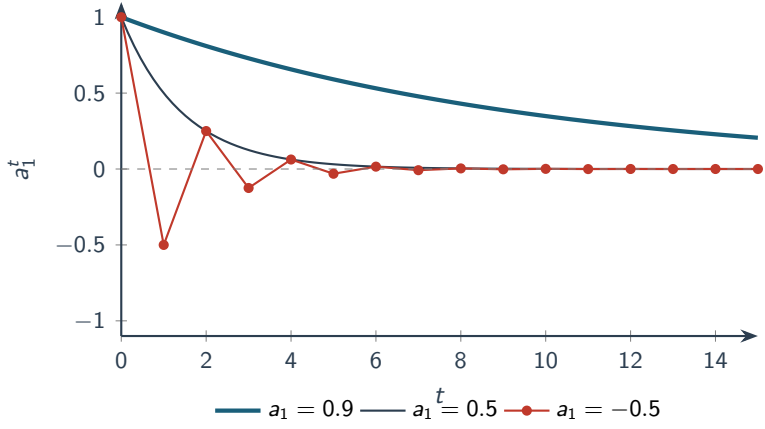
$$y_t = \frac{a_0}{1 - a_1} + \sum_{i=0}^{\infty} a_1^i \, \varepsilon_{t-i}.$$

- $|a_1| < 1$: past shocks decay; the series is convergent (stable). Convergence is direct if $0 < a_1 < 1$, oscillatory if $-1 < a_1 < 0$.
- $a_1 = 1$: the random walk case. The steps above break down (the sum $\sum a_1^i$ diverges); each shock has a **permanent** effect: $y_t = y_0 + a_0 t + \sum_{i=1}^{t} \varepsilon_i$.
- $|a_1| > 1$: the term $a_1^{m+1} y_{t-m-1}$ no longer vanishes — shocks are amplified and the series explodes.

This preview of stability is the central theme — and it generalizes via characteristic roots.

Iteration and Stability: The Role of a_1

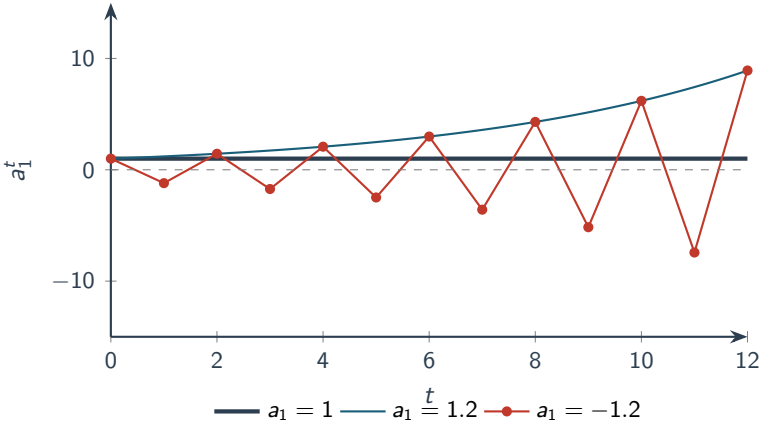
The homogeneous solution a_1^t governs everything. Its behavior depends entirely on a_1:



All three converge because $|a_1| < 1$. Smaller $|a_1| \Rightarrow$ faster decay; a *negative* a_1 makes the path alternate in sign.

When Sequences Do Not Converge

Once $|a_1| \geq 1$, the past no longer fades:



At $a_1 = 1$ a shock has a **permanent** effect (the flat line); for $|a_1| > 1$ shocks are amplified and the series diverges.

The General Solution: Homogeneous + Particular

Iteration breaks down for higher-order equations. A more powerful approach splits the problem in two.

- **Homogeneous part.** Drop a_0 and all shocks, leaving $y_t = a_1 y_{t-1}$. Try a solution of the form $y_t^h = A\alpha^t$; substituting,

$$A\alpha^t = a_1 A\alpha^{t-1} \implies \alpha = a_1,$$

  so $y_t^h = A\, a_1^t$ solves it for *any* constant A. It describes the *disequilibrium dynamics*.
- **Particular part.** Any single solution y_t^p satisfying the full equation — it carries the effect of the forcing process.

The Four-Step Method

1. Form the homogeneous equation and find *all* n homogeneous solutions.
2. Find a particular solution y_t^p.
3. General solution = particular + linear combination of homogeneous solutions.
4. Impose n initial conditions to pin down the arbitrary constants.

Homogeneous Solutions: The Characteristic Equation

Consider the second-order homogeneous equation

$$y_t - a_1 y_{t-1} - a_2 y_{t-2} = 0.$$

Guided by the first-order case, try the solution $y_t^h = A\alpha^t$. Substituting and dividing by $A\alpha^{t-2}$ gives the **characteristic equation**

$$\alpha^2 - a_1\alpha - a_2 = 0,$$

whose two **characteristic roots** are

$$\alpha_1, \alpha_2 = \frac{a_1 \pm \sqrt{a_1^2 + 4a_2}}{2} = \frac{a_1 \pm \sqrt{d}}{2}, \qquad d \equiv a_1^2 + 4a_2.$$

- Each root gives a valid solution, and so does any linear combination:

$$y_t^h = A_1(\alpha_1)^t + A_2(\alpha_2)^t.$$

- The nature of the solution depends on the sign of the discriminant d.

The Three Cases

Case 1 — $d > 0$: two **real, distinct** roots.

$$y_t^h = A_1(\alpha_1)^t + A_2(\alpha_2)^t.$$

Converges iff *both* roots are less than one in absolute value.

Case 2 — $d = 0$: a **repeated** root $\alpha = a_1/2$. A second, independent solution is $t(a_1/2)^t$:

$$y_t^h = A_1(a_1/2)^t + A_2\, t\,(a_1/2)^t.$$

Converges iff $|a_1| < 2$ (the factor t is dominated by the decay).

Case 3 — $d < 0$: **complex** roots (requires $a_2 < 0$). In polar form,

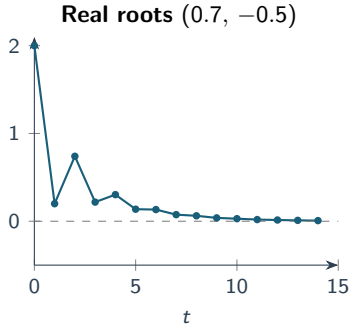
$$y_t^h = \beta_1\, r^t \cos(\theta t + \beta_2), \quad r = \sqrt{-a_2}, \quad \cos\theta = \frac{a_1}{2\sqrt{-a_2}}.$$

Oscillates with frequency θ; converges iff $r = \sqrt{-a_2} < 1$, i.e. $|a_2| < 1$.

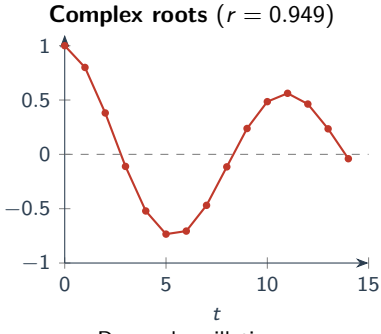
The Three Cases: Examples

- **Real, distinct.** $y_t = 0.2y_{t-1} + 0.35y_{t-2}$ has $d = 1.44 > 0$ and roots $\alpha_1 = 0.7$, $\alpha_2 = -0.5$:
$$y_t^h = A_1(0.7)^t + A_2(-0.5)^t.$$
Both $|\alpha| < 1$, so convergent — but not monotone, because of $(-0.5)^t$.
- **Explosive.** $y_t = 0.7y_{t-1} + 0.35y_{t-2}$ has roots 1.037 and -0.337; the root above one makes $\{y_t\}$ explode.
- **Complex.** $y_t = 1.6y_{t-1} - 0.9y_{t-2}$ has $d < 0$; with $r = \sqrt{0.9} = 0.949$ and $\cos\theta = 0.843 \Rightarrow \theta = 0.567$:
$$y_t^h = \beta_1(0.949)^t \cos(0.567\, t + \beta_2).$$
Damped oscillations, since $r < 1$.

Second-Order Dynamics: Two Contrasting Cases



Convergent, non-monotone



Damped oscillations

Left: $y_t^h = (0.7)^t + (-0.5)^t$ — decays, but the negative root spoils monotonicity. Right: $y_t^h = (0.949)^t \cos(0.567\, t)$ — $r < 1$ gives a wave that dies out. Both are stable because all roots lie inside the unit circle.

Stability and the Unit Circle

Whether real or complex, the roots live in the complex plane. A single condition unifies all cases:

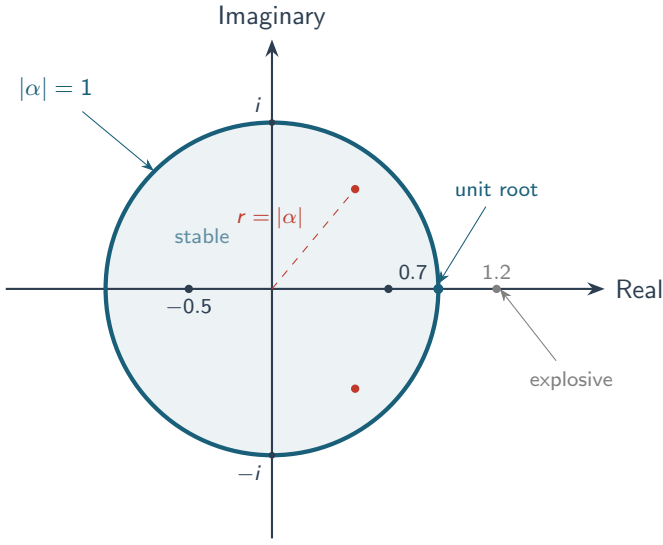
Stability Condition

A difference equation is **stable** (convergent) if and only if
all characteristic roots lie inside the unit circle ($|\alpha_i| < 1$).

- Real roots lie on the horizontal axis; complex roots come in conjugate pairs at distance $r = \sqrt{-a_2}$ from the origin. Stability requires $r < 1$.
- If a root *equals* one, the series has a **unit root** — shocks are permanent (as in the random walk).
- If any root lies *outside* the circle, the series explodes.

This links directly to stationarity in the next course — convergent $\Leftrightarrow$ stationary.

Stability and the Unit Circle



Higher-Order Systems: Stability Shortcuts

For the nth-order equation $y_t = a_0 + \sum_{i=1}^{n} a_i y_{t-i} + x_t$, the n characteristic roots solve

$$\alpha^n - a_1\alpha^{n-1} - a_2\alpha^{n-2} - \ldots - a_n = 0.$$

Rather than compute them directly, useful checks are:

- **Necessary** for all roots inside the unit circle:

$$\sum_{i=1}^{n} a_i < 1.$$

- **Sufficient** for all roots inside the unit circle:

$$\sum_{i=1}^{n} |a_i| < 1.$$

- There is **at least one unit root** if

$$\sum_{i=1}^{n} a_i = 1.$$

Particular Solutions: Deterministic Forcing

The particular solution y_t^p carries the effect of the forcing process. Its form depends on $\{x_t\}$.

- **Constant case ($x_t = 0$).** Try $y_t = c$ (a constant). Then

$$y_t^p = \frac{a_0}{1 - a_1 - a_2 - \ldots - a_n},$$

  valid as long as $\sum a_i \neq 1$. This is the **long-run equilibrium** value.
- **Unit-root case ($\sum a_i = 1$).** The constant fails (denominator is zero). Try a **time trend** $y_t^p = ct$; a unit root injects a deterministic trend into the solution.
- **Time-trend forcing ($x_t = bt$).** Try $y_t^p = c_0 + c_1 t$ and match coefficients.

The recurring trick: if a root equals one, multiply the trial solution by t.

The Method of Undetermined Coefficients

Key insight: linear equations have linear solutions. So the particular solution must be a linear function of a constant, time, and the shocks $\{\varepsilon_t\}$.

- Posit a **challenge solution** containing every term that could plausibly appear, with unknown coefficients.
- Substitute it into the difference equation and choose the coefficients that make the result an **identity** for all t and all $\{\varepsilon_t\}$.
- If no choice works, the challenge form is wrong — revise and repeat.

For $y_t = a_0 + a_1 y_{t-1} + \varepsilon_t$, try

$$y_t = b_0 + b_1 t + \sum_{i=0}^{\infty} \alpha_i \varepsilon_{t-i}.$$

Method of Undetermined Coefficients: First-Order

Substituting the challenge solution into $y_t = a_0 + a_1 y_{t-1} + \varepsilon_t$ and matching coefficients term by term:

- **Shock coefficients:** $\alpha_0 = 1$, $\alpha_1 = a_1$, $\alpha_2 = a_1^2, \ldots$, so $\alpha_i = a_1^i$.
- **Constant and trend:** if $a_1 \neq 1$, then $b_1 = 0$ and $b_0 = a_0/(1 - a_1)$.

The particular solution is therefore

$$y_t^p = \frac{a_0}{1 - a_1} + \sum_{i=0}^{\infty} a_1^i \, \varepsilon_{t-i},$$

exactly the solution found earlier by iteration. The general solution adds the homogeneous term:

$$y_t = \frac{a_0}{1 - a_1} + \sum_{i=0}^{\infty} a_1^i \, \varepsilon_{t-i} + A \, a_1^t.$$

- If instead $a_1 = 1$, the trend survives ($b_1 = a_0$), reproducing the random-walk-with-drift solution.

A Solved Second-Order Problem

Solve $y_t = 3 + 0.9y_{t-1} - 0.2y_{t-2} + \varepsilon_t$.

1. **Homogeneous.** Roots of $\alpha^2 - 0.9\alpha + 0.2 = 0$ are 0.5 and 0.4:

$$y_t^h = A_1(0.5)^t + A_2(0.4)^t.$$

2. **Deterministic particular.** A constant c gives $c = 3 + 0.9c - 0.2c \Rightarrow y_t^p = 10$.
3. **Stochastic particular.** Challenge $\sum_i \alpha_i \varepsilon_{t-i}$ gives $\alpha_0 = 1$, $\alpha_1 = 0.9$, and for $i \geq 2$,

$$\alpha_i = 0.9\,\alpha_{i-1} - 0.2\,\alpha_{i-2} \implies \alpha_i = 5(0.5)^i - 4(0.4)^i.$$

The general solution combines all pieces:

$$y_t = 10 + A_1(0.5)^t + A_2(0.4)^t + \sum_{i=0}^{\infty} \alpha_i\, \varepsilon_{t-i},$$

with A_1, A_2 fixed by two initial conditions.

Lag Operators

When we don't need the explicit coefficients, **lag operators** give a compact alternative. Define

$$L^i y_t \equiv y_{t-i}.$$

Useful properties:

- The lag of a constant is itself: $Lc = c$.
- Powers add: $L^i L^j y_t = L^{i+j} y_t = y_{t-i-j}$, and $L^0 y_t = y_t$.
- Negative powers lead: $L^{-i} y_t = y_{t+i}$.
- **Geometric expansion** (for $|a| < 1$):

$$\frac{1}{1-aL} y_t = \left(1 + aL + a^2 L^2 + \ldots\right) y_t = \sum_{i=0}^{\infty} a^i y_{t-i}.$$

Solving with Lag Operators

Write $y_t = a_0 + a_1 y_{t-1} + \varepsilon_t$ as $(1 - a_1 L) y_t = a_0 + \varepsilon_t$, so

$$y_t = \frac{a_0 + \varepsilon_t}{1 - a_1 L}.$$

Applying the properties ($L a_0 = a_0$, then the geometric expansion):

$$y_t = \frac{a_0}{1 - a_1} + \sum_{i=0}^{\infty} a_1^i \, \varepsilon_{t-i},$$

the same particular solution as before, obtained in two lines.

- **Compact notation.** The general pth-order model becomes

$$A(L)\, y_t = a_0 + B(L)\, \varepsilon_t, \qquad y_t^p = \frac{a_0}{A(L)} + \frac{B(L)}{A(L)}\, \varepsilon_t.$$

- **Stability, restated.** The roots of $1 - a_1 L - \ldots - a_p L^p = 0$ must lie *outside* the unit circle — equivalently, the characteristic roots lie *inside* it.

Bringing It Together: The AR Model

The **autoregressive** model — the workhorse of time-series econometrics — is nothing other than a stochastic difference equation. The AR(p) model is

$$y_t = a_0 + a_1 y_{t-1} + a_2 y_{t-2} + \ldots + a_p y_{t-p} + \varepsilon_t,$$

which is exactly the pth-order linear difference equation with forcing process $x_t = \varepsilon_t$.

- **AR(1):** $y_t = a_0 + a_1 y_{t-1} + \varepsilon_t$. We solved this: $y_t = \dfrac{a_0}{1 - a_1} + \sum_{i=0}^{\infty} a_1^i \varepsilon_{t-i}$.
- The particular solution is the **moving-average representation** — the series written as a weighted sum of past shocks.
- The homogeneous root a_1 governs **persistence**: how long a shock lingers.

Everything from this lecture — solutions, roots, stability — applies directly to AR models.

Stability, Stationarity, and What Comes Next

The stability condition you learned is precisely the **stationarity** condition for AR models.

- $|a_1| < 1$ **(AR(1)) / all roots inside the unit circle (AR(p)):** shocks decay, the series reverts to its mean $a_0/(1 - a_1)$ — the process is **stationary**.
- $a_1 = 1$**:** a unit root — the **random walk**. Shocks are permanent; the series does not revert. This is a **non-stationary** process.
- $|a_1| > 1$**:** explosive — rarely seen in economic data.

The Bridge to the Main Course

Testing whether a series has a **unit root** — stationary or not — is one of the central questions of applied time-series econometrics, and the starting point of Advanced Econometrics.

Application: The Cobweb Model

A classic model of agricultural price volatility. Farmers set output at planting time using *last period's* price (naive expectations).

$$\begin{aligned} \text{Demand:} \quad & d_t = a - \gamma p_t && (\gamma > 0) \\ \text{Supply:} \quad & s_t = b + \beta p_t^* + \varepsilon_t && (\beta > 0) \\ \text{Clearing:} \quad & s_t = d_t, \qquad p_t^* = p_{t-1} \end{aligned}$$

Combining the four equations yields a **first-order stochastic difference equation** in price:

$$p_t = -\frac{\beta}{\gamma}\, p_{t-1} + \frac{a-b}{\gamma} - \frac{\varepsilon_t}{\gamma}.$$

- This has the familiar form $p_t = a_0 + a_1 p_{t-1} + e_t$ with $a_1 = -\beta/\gamma$.
- The negative slope coefficient is what generates the cobweb's **oscillations**.

The Cobweb Model: Solution and Stability

Applying the four-step method:

- **Homogeneous:** $p_t^h = A(-\beta/\gamma)^t$.
- **Particular** (for $\beta/\gamma < 1$): a long-run price plus decaying shocks,

$$p_t^p = \frac{a-b}{\gamma+\beta} - \frac{1}{\gamma}\sum_{i=0}^{\infty}\left(-\frac{\beta}{\gamma}\right)^i \varepsilon_{t-i}.$$

- **General solution** (initial condition p_0):

$$p_t = \frac{a-b}{\gamma+\beta} + \left(-\frac{\beta}{\gamma}\right)^t \left(p_0 - \frac{a-b}{\gamma+\beta}\right) + (\text{shocks}).$$

Stability Condition

The system converges iff $\beta/\gamma < 1$ — the **supply curve must be steeper than demand**.

Prices oscillate around $\dfrac{a-b}{\gamma+\beta}$, dampening if $\beta < \gamma$, exploding if $\beta > \gamma$.

The Cobweb Model: Impulse Responses

How does a supply shock ε_t ripple through prices? The **impact multiplier** is

$$\frac{\partial p_t}{\partial \varepsilon_t} = -\frac{1}{\gamma},$$

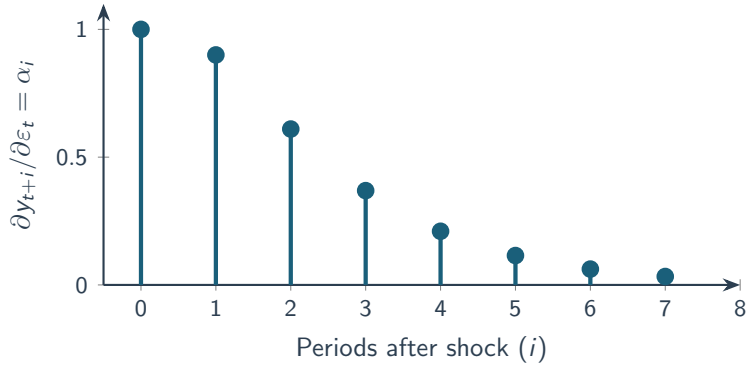
and the effect n periods later is

$$\frac{\partial p_{t+n}}{\partial \varepsilon_t} = -\frac{1}{\gamma}\left(-\frac{\beta}{\gamma}\right)^n.$$

- The sequence of these derivatives is the **impulse response function** — it traces the full time path of a one-time shock.
- With $\beta/\gamma < 1$, the responses **decay and alternate in sign**: a shock's influence is short-lived and oscillatory.
- Impulse responses are central to modern time-series analysis (money-supply shocks, productivity shocks on GDP, ...).

The Impulse Response Function

How does a single shock ε_t propagate? Plot its effect over time — here the coefficients α_i of $y_t = 3 + 0.9y_{t-1} - 0.2y_{t-2} + \varepsilon_t$:



A one-unit shock raises y by 1 on impact, then the effect **decays geometrically** back to zero (because both roots are inside the unit circle). This decaying profile is the **impulse response function** — the central object of applied time-series analysis.

References

[1] Enders, W. (2014). *Applied Econometric Time Series* (4th ed.). Wiley.

[2] Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press.